fn make_row_by_row

Michael Shoemate

Proves soundness of `make_row_by_row` in `mod.rs` at commit f5bb719 (outdated[1]). This constructor is a special case of `make_row_by_row_fallible`. The proof for this constructor appeals to the proof for `make_row_by_row_fallible`.

`make_row_by_row` returns a Transformation that applies a user-specified function to each record in the input dataset.

1 Hoare Triple

Precondition

Compiler-verified

- Generic `DI` (input domain) is a type with trait `RowByRowDomain<DO>`.
- Generic `DO` (output domain) is a type with trait `DatasetDomain`. `DatasetDomain` is used to define the type of the row domain.
- Generic `M` (metric) is a type with trait `DatasetMetric`. `DatasetMetric` is used to restrict the set of valid metrics to those which measure distances between datasets.
- `MetricSpace` is implemented for `(DI, M)`. Therefore `M` is a valid metric on `DI`.
- `MetricSpace` is implemented for `(DO, M)`.
- Argument `input_domain` is of type `DI`
- Argument `input_metric` is of type `M`
- Argument `output_row_domain` is of type `DI::ElementDomain`, as defined by `DatasetDomain`.
- Argument `row_function` is an immutable thread-safe function taking in a value of the carrier type of the element domain associated with `input_domain`, and returning a value of the carrier type of `output_row_domain` or an error.

User-verified

- `row_function` has no side-effects.
- If the input to `row_function` is a member of `input_domain`'s element domain, then the output is a member of `output_row_domain`.

[1] See new changes with `git diff f5bb719..5e43cbf0 rust/src/transformations/manipulation/mod.rs`

Pseudocode

```python
def make_row_by_row(
    input_domain: DI,
    input_metric: M,
    output_row_domain: DO_RowDomain,
    # a function from input domain's row type to output domain's row type
    row_function: Callable([[DI_RowDomain_Carrier], DO_RowDomain_Carrier])
) -> Transformation:
    # Python doesn't model errors in the type system,
    # so in the Rust code the return value from the function is wrapped in Ok
    return make_row_by_row_fallible(
        input_domain, input_metric, output_row_domain, row_function
    )
```

Postcondition

Theorem 1.1. For every setting of the input parameters (`input_domain, input_metric, output_row_domain, outprow_function, DI, DO, M`) to `make_row_by_row` such that the given preconditions hold, `make_row_by_row` raises an error (at compile time or run time) or returns a valid transformation. A valid transformation has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Appropriate output domain). For every member x in `input_domain`, `function`(x) is in `output_domain` or raises a data-independent runtime error.

3. (Stability guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
   `output_metric`, if x, x' are `d_in`-close under `input_metric`, `stability_map`(`d_in`) does not raise an error, and `stability_map`(`d_in`) = `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_metric`.

2 Proofs

Proof. **(Part 1 – appropriate output domain).** Since the preconditions for this constructor are a superset of the preconditions on `make_row_by_row_fallible`, the proof of `make_row_by_row_fallible` applies. Thus, by the output domain proof on `make_row_by_row_fallible`, for all settings of input arguments, the function returns a dataset in the output domain. □

Proof. **(Part 2 – stability map).** The proof of `make_row_by_row_fallible` similarly applies. Thus, by the stability map proof on `make_row_by_row_fallible`, for all settings of input arguments, where x, x' are `d_in`-close under `input_metric`, `function`(x), `function`(x') are `d_out`-close under `output_metric`. □